

Name:

MyDesign Portfolio

Element G Initial Template - AT - 12 points

Element G: Construction of a Testable Prototype

Final Prototype Iteration Explanation (G1)

Prototype Construction of Design #1 (make as many copies, updating the design number, as needed)

Date: **3/26/2025**

Details about all steps of construction (**add step numbers as needed**), including the tools and manufacturing methods used:

	Details	Tools and Manufacturing Methods used
Step 1	Grab the PVC Pipe, measure it to know where you will be cutting the PVC pipe to the desired length of 17 in. and mark.	• PVC Pipe • Ruler • Marker
Step 2	Use the jigsaw to begin cutting the PVC pipe. Cut to desired length using the marks from the previous step.	• Jigsaw • PVC Pipe
Step 3	Sand down the spots where you	• Sandpaper • PVC Pipe

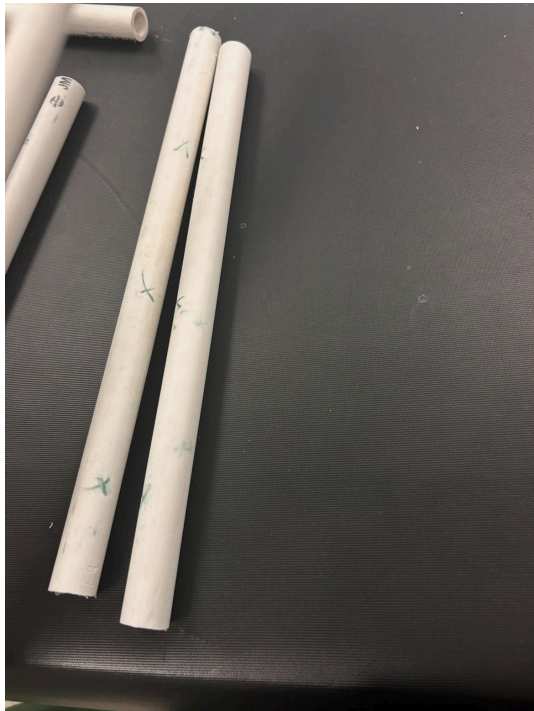
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	cut using sandpaper.	
Step 4	Use the marker and draw "x's" on every spot you will drill into (every 4 in.)	• PVC Pipe • Marker • Ruler
Step 5	Drill on the marked spots of the PVC pipes using a drill with 4mm drill bits. DO NOT DRILL ALL THE WAY THROUGH.	• 4mm drill bits • PVC Pipe • Drill

Photos from the construction process:

***sorry I missed getting picture of when the PVC pipes where being cut**

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The marked and cut PVC pipes
pipes



| Drilled in holes in the marked PVC

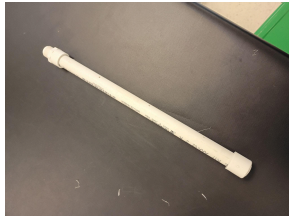


Drilling in the holes into the PVC Pipe

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Final Prototype Construction of Design #2 (change this number to match the design number you're on)

Date:



	Details	Tools and Manufacturing Methods used
Step 1	Add hose connector to pipe	Hose connector
Step 2	Add cap to pipe	Pipe cap

Material name	Description	Where material was purchased	Cost per unit	# units	Total cost per item
Hose connector	Allows hose to connect to pipe	Home Depot	\$0.70	1	\$0.70
Pipe cap	Blocks 1 end of pipe	Home Depot	\$0.70	1	\$0.70

Detailed sketches, which might include CAD if you have learned it in your class:

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Construction of the Final Prototype (G2)

When you've finished construction of your final prototype, you must be sure that your prototype allows sufficient functionality that you could collect objective, measurable data for all or nearly all of your design requirements. This requires you to look back at your design requirements from Element C.

Copy all of your design requirements from Element C into this table and then add how your prototype will allow you to collect objective, measurable data. Note any weaknesses you find or details that will require expert review because they cannot be mathematically modeled or tested,

For example, if you have a design requirement that your design can "transport medicine a distance of 5 meters in 1 minute", is your prototype sufficiently functional such that you can test that? (e.g. can the prototype move the medicine at least a little bit?)

Design Requirement	How will this prototype allow you to collect objective, measurable data?
Even flow of water	We can place buckets under prototype. We can run water through prototype and see if it waters evenly
Doesn't leak	We can run water through prototype and see if it leaks.
Easy to maintain	Run 1 part bleach 10 part water through prototype

Testing or Modeling of Attributes (G3, G4) and Addressing Non-Testable Attributes (G4)

Describe all of your prototype's attributes (features, aspects, details, or subsystems). Demonstrate that your prototype is constructed with sufficient functionality that your prototype can test all attributes of your design. This requires you to look back at your initial proposed design details from Element D.

For example, if your initial design details called for wheels and motors, is your prototype sufficiently functional such that you can test out those wheels and motors?

If there are aspects of your design that you believe cannot be physically prototyped or mathematically modeled within the scope and limits of your class, you must later provide evidence that an expert has reviewed your justifications and that you have incorporated

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their feedback. You do not need to conduct that review in Element G. For now, enter "expert review required" for these attributes in the final column.

Attribute	Description	Describe your prototype's functionality to test this attribute
lightweight	Lighter than 1 pound	Is lighter than 1 pound
Low cost	Cheaper than \$10	Is cheaper than \$10
anti-mold	Doesn't get moldy	Doesn't get monldy
Low maintenance	Maintenance is easy	MAintenance is easy

Addressing Non-Testable Attributes (G4)

Attribute	Justification for why this attribute cannot be modeled or tested with your prototype
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Element G: Constructio n of a Testable Prototype						
	5	4	3	2	1	0
1. Final Prototype Iteration Explanation. The Final Prototype Iteration is _____.	clearly and fully explained	clearly and adequately explained	adequately explained	somewhat explained but with gaps	only minimally explained	is unclear or is missing altogether



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2. Construction of the Final Prototype. The Final Prototype Iteration is _____.	constructed with enough detail to assure that objective data on <i>all</i> or nearly all design requirements could be determined	constructed with enough detail to assure that objective data on <i>many or nearly all</i> design requirements could be determined	constructed with enough detail to assure that objective data on <i>some</i> design requirements could be determined	constructed with enough detail to assure that objective data on at least <i>a few</i> design requirements could be determined	<i>not</i> constructed with enough detail to assure that objective data on at least <i>one</i> design requirement could be determined	<i>not applicable</i> to any design requirements and/or functional claims
3. Testing or Modeling of Attributes. _____ attributes (sub-systems) of the unique solution that can be tested or modeled mathematically are addressed.	<i>All</i>	<i>Most</i>	<i>Some</i>	<i>A Few</i>	<i>No more than one</i>	<i>None of the</i>
4. Addressing non-Testable Attributes. _____ justification is provided for those that cannot be tested or modeled mathematically and thus require expert review.	<i>A well supported</i>	<i>A generally supported</i>	<i>An adequately supported</i>	<i>Developing</i>	<i>Insufficient</i>	<i>No</i>